Network Position as Confounder: Positional Debiasing and Variance-Weighted Learning for Graph-Based Uplift Modeling

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Abstract

Graph-based uplift modeling drives targeted recommendation on social networks, where a user's response to an intervention is shaped by their network position as much as by their own features. Existing GNN-based ITE estimators each introduce a monolithic architecture without articulating which error component its structural inductive bias corrects, and as a result their improvements cannot compose. Graph ITE error appears to mix distinct failure modes—community-biased treatment assignment, position-sensitive potential outcomes, and position-driven heteroscedastic noise—yet we prove that all three collapse into an exact two-term decomposition of PEHE under one latent cause: a node's structural position, a *dual-root confounder* that simultaneously biases treatment assignment and drives heteroscedastic outcome noise. The decomposition prescribes two architecture-agnostic plug-ins: **Graph Positional Encoding (GPE)** injects node2vec embeddings via cross-attention to reduce the bias term, and **Variance-Weighted Learning (VWL)** reweights the loss by estimated per-sample noise to reduce the variance term. Across seven base models and ten datasets, GPE cuts mean PEHE by 8.6% over six separate-head backbones at Wilcoxon $p < 10^{-5}$ and wins 93% of cells in the predicted moderate-noise regime, while VWL delivers a +303% relative Qini lift on continuous-outcome Hillstrom spend with preliminary binary-benchmark gains; the two modules compose orthogonally as Prop. 1(c) predicts—VWL alone cuts GDC's PEHE by up to 29% on variance-dominated cells and joint GPE+VWL by up to 20% on bias-dominated cells.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Neural networks*; • **Mathematics of computing** → *Probabilistic inference*.

Keywords

uplift modeling, individual treatment effect, graph neural networks, positional encoding, heteroscedasticity, causal inference

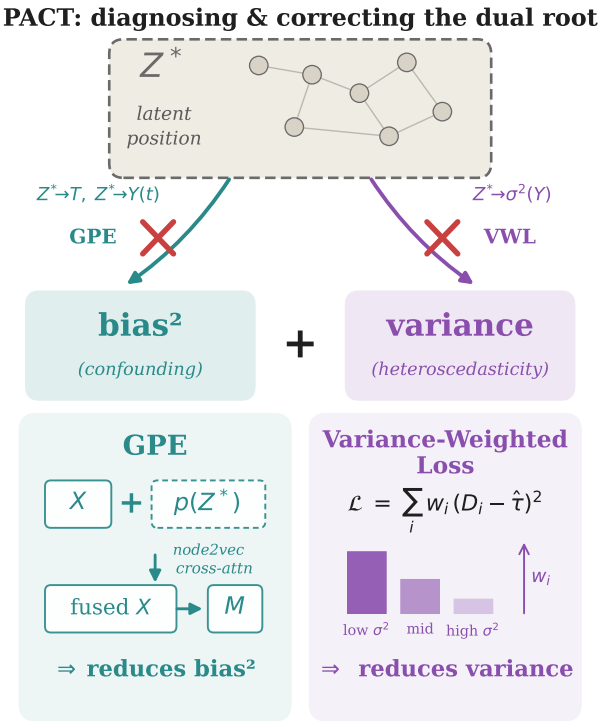

Figure 1: Dual-root confounder and its two plug-in corrections. Z^* drives both the bias pathway ($Z^* \to T, Y(t)$) and the heteroscedasticity pathway ($Z^* \to \sigma^2(Y)$); Prop. 1 decomposes PEHE accordingly. GPE proxies Z^* at the input; VWL reweights the loss by $1/\hat{\sigma}_t^2$. Both attach to any backbone M.

1 Introduction

Targeted interventions are only as valuable as the models that predict who will respond to them. Digital platforms invest heavily in personalized incentives [11, 26], and a misallocated promotion is not merely wasteful—it degrades user experience and the signal-quality of future recommendations. The core modeling task is estimating the Individual Treatment Effect (ITE): distinguishing users who genuinely change behavior under treatment from those who would have acted identically regardless. On social platforms, this task acquires an extra dimension through network interference: a user's response is shaped not only by their own features but by the concurrent actions of their social connections [3]. Users in tightly-knit communities respond more strongly to identical promotions than socially isolated users [10, 22]—a gap that i.i.d. estimators [18, 20, 30] cannot recover. Figure 1 illustrates the phenomenon: treatment propensity and outcome-noise variance are both spatially structured in real social networks, co-varying with each node's position in the graph.

A growing line of GNN-based ITE methods has responded to this challenge—NetDeconf [13] (2020), GIAL [8], GNUM [40], and the current SOTA GDC [14] (2025)—yet every entry in the sequence introduces a monolithic architecture whose improvements cannot compose with one another. More importantly, the field advances without a principled account of *which* error source graph information is correcting: is there one, or several? Evaluation compounds the issue. Real-world graph uplift data at recommendation scale is effectively unavailable, so the literature has consolidated around two semi-synthetic benchmarks—BlogCatalog and Flickr—that graft simulated treatments and outcomes onto real graphs. The combination of monolithic designs and a thin, semi-synthetic benchmark suite leaves the community unable to decide which model components matter for which reasons.

Our primary contribution is diagnostic and theoretical. We identify and formally characterize a node's latent structural position Z^*—its community membership, centrality, and bridging role—as a *dual-root confounder*: Z^* simultaneously acts along (1) $Z^* \to T$ (position influences treatment assignment), (2) $Z^* \to Y(t)$ (position influences potential outcomes), and (3) $Z^* \to \sigma^2(Y)$ (position drives outcome-noise variance). The first two paths constitute classical confounding; the third drives heteroscedasticity. We prove (Prop. 1) that every graph ITE model's estimation error reduces to just two PEHE components—a bias term from paths (1–2), a variance term from path (3)—both driven by the *same* latent Z^*. The reduction is surprising because prior graph-ITE work attacks each apparent symptom (community bias, position-sensitive outcomes, heteroscedastic noise) as if they were separate problems; unification under Z^* is what makes a single modular fix possible.

How our contribution differs from NetDeconf, GIAL, GNUM, GDC. Every prior graph-ITE method attacks only the bias pathways: NetDeconf balances treated/control representations, GIAL uses adversarial discriminators, GNUM applies transformed targets, GDC disentangles treatment-conditioned aggregation—all different strategies for correcting confounding from $Z^* \to T$ and $Z^* \to Y(t)$. *None of them addresses path (3) ($Z^* \to \sigma^2$).* Our paper is the first to name this third pathway, prove it induces a second PEHE error term under the same latent cause, and prescribe a direct correction for it (Variance-Weighted Learning). VWL is therefore *orthogonal* to all prior methods rather than competing with them—it plugs into NetDeconf, GIAL, GDC, or a plain TARNet with equal ease (empirically: adding VWL to GDC, the current SOTA, yields a further 29% PEHE reduction on PubMed at ρ=30). The contribution is not a better bias correction; it is (a) a unifying diagnostic explaining why prior methods all work and what they are missing, (b) the missing variance correction, and (c) a modular recipe letting practitioners keep their existing backbone and add the two plug-ins instead of replacing the architecture.

Graph Positional Encoding (GPE) approximates Z^* through node2vec embeddings and fuses them into an existing backbone via cross-attention, implementing Veitch et al.'s proxy-confounder adjustment [34] to reduce the bias term. **Variance-Weighted Learning** estimates per-sample noise via a $\log \sigma^2$ head and reweights the training loss following Gauss–Markov efficiency [2] to reduce the variance term. GPE operates at the *input* level and variance weighting at the *loss* level, so neither touches the base model's internals; both are plug-in modules that compose with any graph ITE backbone. The prescriptions are complementary in a way the theory predicts and the experiments confirm: at high noise, GPE alone can amplify position-correlated variance, while the combination addresses both error components.

Because large-scale real-world graph uplift data does not exist, we validate the theory through a controlled plug-in protocol on seven base models—spanning NetDeconf [13] to GDC [14]—over five graph datasets, including the real-graph BlogCatalog and Flickr benchmarks from Guo et al. [13] used directly. On five real-world tabular uplift benchmarks we test VWL *as a method*: whether inverse-variance reweighting generalises to non-graph heteroscedasticity sources (user-level revenue variance, treatment-rate imbalance), distinct from the dual-root theory itself (which concerns network position). Extensive controlled runs support the theory: GPE improves mean PEHE on 84% of separate-head cells overall (76/90, Wilcoxon $p < 10^{-5}$), rising to 93% (56/60) on the regime where Proposition 1(a) predicts gains—moderate-noise cells $\rho \leq 20$ across the five T-learner backbones (TARNet, GDC, NetDeconf, GIAL, GNUM). All six separate-head backbones benefit, with per-backbone gain graded by bias-pathway headroom (Prop. 1(a)): X-learner's cross-fitted pseudo-outcome gives it the least headroom through μ_{1-t} and correspondingly the smallest gain (−3.4%), exactly as the decomposition predicts. VWL alone cuts GDC's error by up to 29% on variance-dominated cells while joint GPE+VWL wins on bias-dominated cells (up to 20% on GDC at high noise), and variance weighting delivers +303% relative Qini on Hillstrom spend (4×) in its continuous-outcome design regime, with preliminary binary-benchmark gains of +129% on Hillstrom-visit and +16% on Criteo (reported with seed-variance caveats)—the three tabular settings with a non-trivial treatment-effect signal. Three supplementary binary benchmarks (X5, Lenta, RetailHero) exhibit $|\Delta\text{Qini}|<0.005$ at every tuned configuration (three base learners × four δ values; see Appendix A), confirming the theory's prediction that variance reweighting cannot refine a near-zero treatment-effect signal. Our contributions are:

- **Theoretical.** We prove that network position is a dual-root confounder and derive a PEHE decomposition that unifies bias and variance under a single latent cause (Section 3.2).
- **Methodological.** We propose two corrections prescribed directly by the decomposition—GPE (input-level, bias) and Variance-Weighted Learning (loss-level, variance)—that compose with any base estimator (Section 4). **VWL is the first graph-ITE module to attack path (3)**; it is orthogonal to every prior bias-correction architecture (NetDeconf, GIAL, GNUM, GDC) rather than a replacement for them.
- **Empirical.** Across 2,000+ controlled runs on seven base models and ten datasets (five graph, five tabular), we validate the theory; code at https://anonymous.4open.science/r/PACT-GL.

2 Related Work

ITE estimation and uplift modeling. Traditional treatment-effect estimation spans tree-based methods [4, 27] and meta-learner frameworks (S-, T-, X-learners [18, 20]); deep variants extend these via representation learning [9, 16, 30]. The theoretical foundation most relevant to our Variance-Weighted Learning module is the R-Learner [25]: reformulating CATE as a residual-on-residual regression and weighting each sample by inverse conditional variance yields quasi-oracle convergence rates. This is the Gauss–Markov–style efficiency argument we import as a loss-level plug-in for any of the base models above—not a replacement for any of them. These methods assume i.i.d. samples and implicitly treat estimation bias and outcome-noise heterogeneity as independent error sources—an assumption we will show is untenable on social platforms, where both trace to a node's network position.

Causal inference under network interference. A focused line of GNN-based ITE estimators addresses the i.i.d. violation above. NetDeconf [13] (2020) pioneered GCN-based deconfounding with Wasserstein representation balancing. GIAL [8] added mutual-information maximization and adversarial balance; GNUM [40] introduced transformed-target estimation; GDC [14] (2025, current

SOTA) disentangles confounder, instrumental, and adjustment features. CauGramer [36] (ICLR 2025) proposes a full causal graph Transformer with minimax moment constraints, requiring complete architecture replacement. Closely adjacent are Ma & Tresp [23] (GNN-based effect estimators with policy regret bounds), Huang et al. [15] (two-channel GCN with masked attention for spillover magnitude), and Zhao et al. [38] (dual-weighting attention for heterogeneous interference); the multi-hop gated aggregation recipe itself goes back to AliNet [32] from the KG-alignment literature. Each work introduces a monolithic design whose innovations do not transfer to other models; more fundamentally, none of these papers names *which* error component its structural inductive bias is actually correcting. We argue that all of them are attacking partial symptoms of a single latent cause—a node's structural position, which simultaneously confounds treatment assignment and drives heteroscedastic outcome noise—and that naming this cause admits modular corrections where the field currently has only monolithic ones. Crucially, all five methods above attack only the *bias* half of this cause; **none of them implements a variance-weighted objective for the heteroscedasticity pathway**. This is precisely the gap Variance-Weighted Learning fills, and why our contribution is additive to (not competing with) these existing architectures.

Positional and structural encoding in GNNs. Standard message-passing GNNs cannot distinguish structurally equivalent nodes that occupy different global positions [35], motivating a rich positional-encoding (PE) literature. Absolute encodings inject topological statistics into attention: Graphormer [37] uses centrality, spatial distance, and edge features. Spectral encodings use Laplacian eigenvectors: SAN [19] reweights attention by eigenvalues, while SignNet [21] handles the sign ambiguity inherent in spectral PE. Hybrid recipes such as GraphGPS [28] combine PE with local message-passing and global attention. In recommendation, PGTR [7] augments LightGCN with spectral, degree, and PageRank encodings for collaborative filtering. Random-walk embeddings such as node2vec [12] offer a lightweight, offline-precomputable alternative that captures both structural equivalence and community membership. None of this work was developed for causal ITE, and none studies positional information as a proxy for an unobserved structural confounder—the role that our GPE plays by fusing node2vec embeddings into *existing* causal GNN backbones rather than replacing them.

Heteroscedastic modeling in deep learning and ITE. Modeling input-dependent noise variance in neural networks dates to Nix & Weigend (1994) and was revisited in the multi-task uncertainty weighting framework of Kendall & Gal [17]. Within the ITE literature, variance-weighted estimators have been explored for tabular settings [1, 24, 31]. This line of work treats heteroscedasticity as a generic modeling concern independent of confounding; it does not connect variance structure to structural causes, and it has not been adapted to graph-based ITE or to pseudo-outcome regression. Two practical challenges distinguish our setting: the regression target (pseudo-effect D_i) is itself a noisy estimate rather than a direct observation, and binary outcomes admit a qualitatively different variance structure. We address both, via a $\log \sigma^2$ parameterization with gradient-isolated cross-fitting for continuous outcomes and an analytical Bernoulli variance $\sigma^2 = \mu(1-\mu)$ for binary outcomes.

Positioning. We bridge these four lines by identifying network position as a *dual-root confounder* and deriving two modular, architecture-agnostic corrections prescribed by the PEHE decomposition it induces. To our knowledge, this is the first controlled plug-in evaluation in graph-based ITE estimation, extending the plug-and-play paradigm that has proven valuable elsewhere in graph recommendation [39].

3 Preliminaries

3.1 Graph ITE and Pseudo-Outcome Regression

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph with adjacency matrix $\mathcal{A}$. Each node i carries features $X_i \in \mathbb{R}^d$, a binary treatment $T_i \in \{0,1\}$, and an observed outcome Y_i; we write $X^i_{\mathcal{N}}$ for aggregated features over multi-hop neighbors $\mathcal{N}_i^{1,\dots,r}$. The dataset is $\mathcal{D} = \{(X_i, T_i, Y_i)\}_{i=1}^n$ paired with $\mathcal{A}$. Because outcomes depend on peers, SUTVA is replaced by three standard assumptions on networked data [13]:

Assumption 1 (Network Consistency). *$Y_i = Y_i(T_i, \mathcal{T}_i)$, where $\mathcal{T}_i = \mathrm{agg}(T_{\mathcal{N}_i^{1,\dots,r}})$ aggregates neighbors' treatments.*

Assumption 2 (Network Unconfoundedness). $Y_i(T_i, \mathcal{T}_i) \perp (T_i, T^i_{\mathcal{N}}) \mid X_i, X^i_{\mathcal{N}}$.

Assumption 3 (Network Overlap). $0 < p(T_i, T^i_{\mathcal{N}} \mid X_i, X^i_{\mathcal{N}}) < 1$.

The estimand is the Network CATE $\tau(X_i) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i, X^i_{\mathcal{N}}]$. Since both potential outcomes are never jointly observed, we adopt the standard pseudo-outcome reduction of Künzel et al. [20] (the X-learner): regress against an imputed target $\tilde{D}$ satisfying $\mathbb{E}[\tilde{D} \mid X] = \tau(X)$, which makes minimizing $\mathbb{E}[(\tilde{D} - \hat{\tau})^2]$ equivalent to minimizing the oracle risk $\mathbb{E}[(\tau - \hat{\tau})^2]$. In practice we fit outcome models $\mu_t = \mathbb{E}[Y \mid T{=}t, X, X_{\mathcal{N}}]$ for $t \in \{0,1\}$, form cross-residuals $\mathrm{D}_1 = Y(1) - \mu_0$ and $\mathrm{D}_0 = \mu_1 - Y(0)$, and combine them via propensity-weighted regression. This construction is an off-the-shelf tool, not a contribution of this work; our contribution is the dual-root identification of network position as a shared cause of the bias and variance terms that any such pseudo-outcome regressor incurs. Every symbol introduced here—$X_i, T_i, Y_i, \mathcal{A}, \mathcal{N}, \mu_t, \tilde{D}$—is used without redefinition in the method section.

3.2 PEHE Decomposition and the Dual-Root Structure

Let $Z^* \in \mathcal{Z}$ denote the latent structural position of a node—a low-dimensional summary of community membership, centrality, and bridging role that is not observed but is partially recoverable from the graph topology. We posit three simultaneous causal paths from Z^*: (i) $Z^* \to T$ (position shifts treatment assignment), (ii) $Z^* \to Y(t)$ (position shifts potential outcomes), and (iii) $Z^* \to \sigma^2(Y)$ (position drives outcome-noise variance). Paths (i)–(ii) together constitute classical confounding; path (iii) drives heteroscedasticity. Network unconfoundedness (Assumption 2) conditions only on observed $(X, X_{\mathcal{N}})$, but Z^* is *residual*—the structural information not captured by raw features—so both effects persist in any estimator trained on X and $X_{\mathcal{N}}$ alone.

What is non-obvious is not the decomposition itself (every MSE admits one) but that on graph ITE the apparently-distinct failure modes—community-biased treatment assignment, position-sensitive potential outcomes, position-driven heteroscedastic

noise—all trace to the *same* latent cause Z^*, so a single proxy strategy and a single inverse-variance strategy jointly suffice. Concretely, the squared PEHE admits the standard decomposition

$$\mathrm{PEHE}^2 = \underbrace{\mathbb{E}\big[(\hat{\tau}(X) - \bar{\tau}(X))^2\big]}_{\text{bias}^2} + \underbrace{\mathbb{E}\big[\mathrm{Var}[\tilde{D} \mid X]\big]}_{\text{variance}}, \tag{1}$$

with $\bar{\tau}(X) = \mathbb{E}[\tau \mid X]$ the best estimator given X. The bias term is inflated whenever Z^* confounds $T \leftarrow Z^* \rightarrow Y(t)$; the variance term is inflated whenever $\mathrm{Var}[\tilde{D} \mid X] = \sigma^2(X, Z^*)$ is position-dependent. The variance term further absorbs first-stage estimation error from $\hat{\mu}_0, \hat{\mu}_1$, which we control with gradient-detached cross-fitting (Algorithm 1). Because the two terms are orthogonal error sources, they call for orthogonal corrections.

Proposition 1 (Network Position as Dual Error Root). *Under network unconfoundedness conditional on $(X, X_{\mathcal{N}}, Z^*)$:*

- *(a)* ***Proxy reduces bias.*** *If $p(Z^*)$ satisfies Veitch et al.'s network unconfoundedness condition [34]—i.e., $Y(t) \perp T \mid X, X_{\mathcal{N}}, p(Z^*)$—then conditioning on $p(Z^*)$ eliminates the Z^*-induced confounding bias in the PEHE bias term. Bias reduction is monotone-nondecreasing in proxy quality: a better approximation of Z^* yields a smaller* or equal *residual bias, with saturation once Z^* is sufficiently proxied (e.g., a position-aware backbone leaves no headroom for an external proxy).*
- *(b)* ***Inverse-variance weighting reduces variance.*** *When $\mathrm{Var}[\varepsilon \mid X] = \sigma^2(X, Z^*)$ is position-dependent, the feasible GLS estimator $\hat{\tau}_{\mathrm{WLS}} = \arg\min \sum_i w_i (D_i - \hat{\tau}(X_i))^2$ with $w_i = 1/\hat{\sigma}^2(X_i)$ achieves $\mathrm{Var}(\hat{\tau}_{\mathrm{WLS}}) \preceq \mathrm{Var}(\hat{\tau}_{\mathrm{OLS}})$ in the Löwner order, directly reducing the variance term in Eq. 1.*
- *(c)* ***Complementarity at high noise.*** *The two corrections act on orthogonal terms of Eq. 1. Let GPE reduce the bias term by factor $\alpha \in [0, 1)$ (i.e.,* $\mathrm{bias}^2 \rightarrow (1 - \alpha)\,\mathrm{bias}^2$*) and, when applied without variance correction, inflate the variance term by factor $(1 + \gamma(\bar{\sigma}^2))$, where $\bar{\sigma}^2 = \mathbb{E}[\sigma^2(X, Z^*)]$ and $\gamma(\cdot)$ is monotone-nondecreasing (the proxy amplifies position-correlated signal in the pseudo-labels $\tilde{D}$, which also carries position-driven noise). Then GPE-alone strictly reduces* PEHE^2 *iff*

$$\alpha \cdot \mathrm{bias}^2 \;>\; \gamma(\bar{\sigma}^2) \cdot \mathbb{E}\big[\mathrm{Var}[\tilde{D} \,|\, X]\big], \tag{2}$$

which fails once heteroscedasticity dominates. VWL (Prop. 1(b)) scales the variance term by factor $(1 - \kappa)$ for $\kappa \in (0, 1]$, restoring the inequality and making the combined correction monotone in both α and κ.

Proposition 1 synthesises Veitch et al.'s proxy-confounder theorem [34], Gauss–Markov efficiency [2], and the bias–variance PEHE decomposition [30] for the graph ITE setting. Two caveats on (b): the Löwner-order guarantee is linear, but for nonlinear backbones the inverse-variance–weighted objective still minimizes a reweighted conditional risk, with the R-learner [25] supplying nonlinear-regime support; and (b) is an oracle statement—we substitute a gradient-detached cross-fitted $\log \sigma^2$ head (§4.2) to avoid first-stage error inflating the weights. Empirically, the decomposition predicts the exact pattern in §5: GPE alone helps at low noise, can hurt in isolation at high noise, and the combination dominates. Part (a) does not require node2vec specifically—any topology-dependent proxy reduces bias, verified against a Graphformer backbone in §6.

4 Method

The dual-root decomposition prescribes two architecture-agnostic corrections: **Graph Positional Encoding** (GPE, §4.1) attacks the bias term by injecting a proxy for Z^* at the input level (Proposition 1(a)); **Variance-Weighted Learning** (§4.2) attacks the variance term by reweighting the pseudo-outcome loss (Proposition 1(b)). **Scope.** Within graph ITE, Prop. 1 establishes Z^* as the unifier of both error terms (§5). Outside graph ITE (tabular, §5.4), heteroscedasticity has other sources (revenue variance, treatment-rate imbalance); VWL applies as a general-purpose inverse-variance correction on D_i regardless of cause. Figure 2 and Algorithm 1 give the full training procedure for an arbitrary base model M.

4.1 Graph Positional Encoding (GPE)

GPE implements the proxy of Proposition 1(a) via node2vec [12] embeddings and multi-head cross-attention. Standard message-passing GNNs—including every base model we evaluate—cannot distinguish two nodes with isomorphic local neighborhoods, so global position is not recoverable from deeper r-hop aggregation alone; GPE injects precisely this missing signal. We fuse the raw feature matrix $h \in \mathbb{R}^{n \times d_h}$ with pre-computed node2vec embeddings $p \in \mathbb{R}^{n \times d_p}$ in a cross-attention block where features are queries and positional embeddings are keys and values:

$$Q = hW_Q,\; K = pW_K,\; V = pW_V, \tag{3}$$

$$\mathrm{head}_k = \mathrm{softmax}\Big(Q_k K_k^\top / \sqrt{d_k}\Big) V_k, \tag{4}$$

$$z = \mathrm{LN}\big([h;\; W_o\,\mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_H)] + f_{\mathrm{res}}([h; p])\big), \tag{5}$$

with $W_{Q,K,V,o}$ learnable projections, LN layer normalization, $[\cdot;\cdot]$ concatenation, f_{res} a residual projection. The output z replaces h as the input to any downstream GNN backbone; because p is offline-fixed, the only runtime cost is the cross-attention block itself.

Why node2vec? Proposition 1(a) only requires $p(\cdot)$ to be a consistent proxy for Z^*, and we verify in §6 that node2vec, Laplacian eigenvectors, and degree+PageRank give comparable PEHE—consistent with the *any-proxy* prediction. We default to node2vec as the cheapest option that jointly captures structural equivalence and community membership without the per-graph eigendecomposition + sign-ambiguity cost of spectral encodings [21].

4.2 Variance-Weighted Learning

Variance-Weighted Learning implements Proposition 1(b) at the loss level. Standard meta-learners weight every sample equally and so implicitly assume homoscedastic noise, which is untenable on networks: some nodes have highly consistent responses while others are erratic, with the variability tracking position. We model the pseudo-outcome from Section 3 as $D_i = \tau(X_i) + \varepsilon_i$ with $\mathrm{Var}[\varepsilon_i \mid X_i] = \sigma^2(X_i)$, and use the identity $\sigma^2 = \mathbb{E}[D^2 \mid X] - \tau^2(X)$ to train a dedicated variance head. To avoid the degeneracy where an unconstrained $\hat{\sigma}^2$ head collapses to zero, the head predicts $\log \sigma^2$ with regression target $\log((D - \hat{\tau})^2 + \epsilon)$, where $\hat{\tau}$ is a pilot estimate

from the outcome heads with gradients detached (cross-fitting—this prevents first-stage error from inflating the weights, consistent with the feasibility caveat in Section 3.2). The sample weights and reweighted regression objective are

$$w_i = \frac{1}{\max\{\hat{\sigma}_t^2(X_i),\ \delta\}}, \qquad \hat{\tau} = \arg\min_{\hat{\tau}'} \sum_i w_i (D_i - \hat{\tau}'(X_i))^2, \quad (6)$$

with $\delta > 0$ a numerical floor. For separate-head backbones (T-learner, X-learner, and the graph causal models in §5) the variance head is treatment-indexed, $\hat{\sigma}_t^2$, matching the convention used in Algorithm 1; we drop the subscript where context is unambiguous.

Binary-Outcome Adaptation. Most recommendation outcomes—click, conversion, purchase—are binary, and the learned $\log \sigma^2$ head becomes unnecessary in this regime. For $Y_t \mid X \sim \text{Bernoulli}(\mu_t(X))$,

$$\sigma_t^2(X_i) \;=\; \text{Var}[Y_t \mid X_i] \;=\; \mu_t(X_i)\big(1 - \mu_t(X_i)\big), \quad (7)$$

so the conditional variance is determined analytically by the outcome head itself. Substituting Eq. 7 into Eq. 6 gives

$$w_i \;=\; \frac{1}{\max\big\{\hat{\mu}_t(X_i)\big(1 - \hat{\mu}_t(X_i)\big),\ \delta\big\}},$$

which requires no extra parameters, cross-fitting, or second backward pass. Near-certain predictions ($\hat{\mu}_t \to 0, 1$) are upweighted and uncertain ones ($\hat{\mu}_t \approx 1/2$) downweighted; the bounded Bernoulli variance ($\leq 1/4$) means this is a small dynamic range, so we expect Bernoulli reweighting to primarily stabilise training rather than shift mean Qini—a pattern confirmed in §5.3.

Remark 1 (Predicted Interaction). *Proposition 1(c) predicts that at high noise ($\rho \geq 30$), GPE alone can be counterproductive: the proxy $p(Z^*)$ amplifies the position-dependent signal, but the same position also drives the noise $\sigma^2(Z^*)$. The model overfits to noisy pseudo-labels. Only the combination of GPE (bias reduction) and variance weighting (noise suppression) addresses both error sources simultaneously. We validate this prediction in Section 5.3.*

4.3 Overall Framework

Figure 2 summarises the two plug-ins: GPE replaces h with fused z on the forward pass; on the backward pass the pseudo-outcome loss is reweighted by Eq. 6, with the $\log \sigma^2$ head trained under gradient-detached cross-fitting. Costs are constant-factor: node2vec is a one-time offline $O(|\mathcal{V}|LW)$ walk cost; GPE adds a cross-attention block ($O(n^2 d)$ full, or $O(n)$ concatenation fusion as an alternative, §6); Variance-Weighted Learning adds one $\log \sigma^2$ MLP of the same size as $\hat{\mu}_t$ and no extra backward pass. The full procedure is given in Algorithm 1; part (c) of Proposition 1 predicts orthogonal composition, validated in §5.3.

5 Experiments

5.1 Experimental Setup

Graph datasets. Three semi-synthetic graph datasets derived from real-world citation networks—CoraFull (19,793 nodes), DBLP (17,716), and PubMed (19,717) [6]—extend the Guo et al. [13] protocol with (i) Louvain community detection [5] with intra-community edge boosting (×3), (ii) multi-hop treatment spillover via normalized adjacency powers up to 5 hops, and

Algorithm 1 Plug-in Training Procedure

Input: Data $\mathcal{D} = \{(X_i, T_i, Y_i)\}_{i=1}^n$, base model M
Parameters: Smoothing δ, epochs K
Compute positional embeddings p via node2vec (offline)
Initialize M's parameters + variance head $\hat{\sigma}^2$
for each epoch $k = 1, \ldots, K$ **do**
  $\tilde{X} \leftarrow \text{GPE}(X, p)$ *// input-level plug-in*
  $\hat{\mu}_t \leftarrow M(\tilde{X}, \mathcal{A})$ *// base model forward*
  Compute residuals $\text{D}_t = (-1)^t [\hat{\mu}_{1-t}(X) - Y]$
  Update $\hat{\sigma}_t^2$ by regressing on $\log((\text{D}_t - \hat{\tau}_t)^2 + \epsilon)$ *// $\hat{\tau}_t$ detached*
  $w_i \leftarrow 1/\max(\hat{\sigma}_t^2(X_i), \delta)$ *// loss-level plug-in*
  Update M via $\nabla \sum_i w_i (\text{D}_i - \hat{\tau}(X_i))^2$
end for

(iii) per-node heteroscedastic noise $\varepsilon_i \sim \mathcal{N}(0, \sigma(Z_i))$ with noise scale derived from SVD-compressed node features projected onto a random direction and linearly mapped to $[-\rho, \rho]$. We sweep $\rho \in \{5, 10, 15, 20, 30\}$ to span the heteroscedasticity range of our DGP. The node-level outcome magnitude in Cora/DBLP/PubMed is on the order of 30, so ρ=30 places noise at roughly signal scale (extreme case where VWL should dominate per Prop. 1(c)), ρ=5 keeps noise well below signal (GPE-dominated regime), and $\{10, 15, 20\}$ sweep the transition. We omit $\rho < 5$ because at that level the additive noise is indistinguishable from feature-driven outcome variance in our DGP; the Guo et al. BlogCatalog/Flickr benchmarks operate in this low-noise regime by their own generation design, and Table 2 plus §5.3 show our corrections still help there. Treatment probability incorporates both self-features and multi-hop neighbor influence: $P(T_i{=}1) = \text{sigmoid}(p_1^i - p_0^i)$, $p_t = \kappa_1 Z^\top c_t + \kappa_2 \sum_{r=1}^h \gamma^r \mathcal{A}^r Z c_t$, with c_0, c_1 SVD-derived representative vectors (full parameters in the supplementary materials).[1] To mitigate DGP-circularity concerns, we also evaluate on the BlogCatalog (5,196 nodes) and Flickr (7,575) benchmarks from Guo et al. [13], whose generation protocol injects no community bias.

Tabular datasets. To validate variance weighting independent of graph topology—(a) isolating VWL from any graph-specific artifact, and (b) probing applicability to production recommendation regimes where graph ITE data is unavailable—we evaluate the three tabular uplift benchmarks covering the regimes where the theory predicts VWL should bite: Hillstrom spend (42K rows, continuous outcome with user-level noise heterogeneity), Hillstrom visit (42K, unbalanced binary, $\mu \approx 15\%$), and Criteo Uplift (1M, balanced binary, 85% treated, strong baseline). Three additional low-baseline binary benchmarks (RetailHero, Lenta, X5) are relegated to the appendix (Appendix A): they exhibit $|\Delta\text{Qini}| < 0.005$ at every tuned configuration, consistent with the theory's prediction that variance reweighting cannot refine a near-zero treatment effect. For the continuous outcome we use the learned $\log \sigma^2$ head; for binary outcomes we use the Bernoulli variance adaptation ($\sigma^2 = \mu(1 - \mu)$).

Base models. We test seven GNN-based ITE estimators: *meta-learners*—BNN [16] (S-learner), TARNet [30] (T-learner), X-learner [20]; *graph causal models*—NetDeconf [13], GIAL [8], GNUM [40], and GDC [14] (current SOTA). The row labelled "PACT" uses X-learner as the separate-head base and applies

[1] Shared parameters: C=5, β_0=−0.3, β_1=0.3, γ=0.6, hops=5.

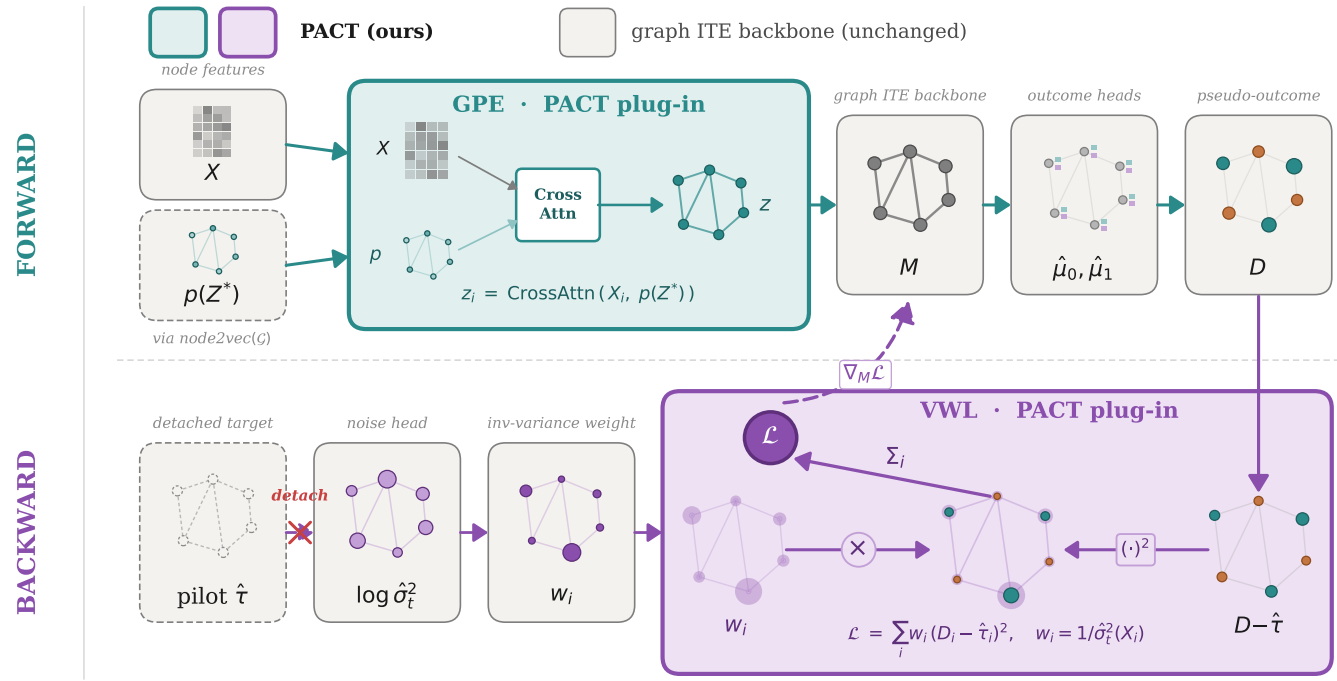

Figure 2: One PACT training step. Forward (top): GPE (teal) fuses X with node2vec proxy $p(Z^*)$ via cross-attention; fused z feeds M. Backward (bottom): VWL (purple) reweights the pseudo-outcome loss by $1/\hat{\sigma}_t^2(X_i)$; pilot $\hat{\tau}$ is gradient-detached (red x).

GPE+Var; the plug-ins compose with every other model and we show the full cross-product in Tables 1–2. All models share a 5-layer GAT backbone ([256, 128, 128, 128, 128], 4 heads) with 4-layer MLP outcome heads (hidden 128). GPE uses 128-dim node2vec embeddings (100 walks, length 30) with 4-head cross-attention at dimension 256.

Training protocol. Adam (lr=10^{-3}, weight decay 10^{-4}, grad-clip 5), 200 epochs, 60/20/20 split, 5 seeds (mean±std), early-stop on validation PEHE (graph) or Qini (tabular). Extended ablations and visualizations are in the supplementary materials at https://anonymous.4open.science/r/PACT-GL.

5.2 GPE as a Modular Plug-in

Table 1 reports rooted PEHE (mean ± std over 5 seeds) across all 15 dataset × noise-level combinations. GPE consistently reduces PEHE across all T-learner and separate-head architectures. BNN, the sole S-learner, shows no improvement—confirming that GPE requires architectures with separate treatment-control prediction pathways to condition on position independently (see below).

GPE's benefit is largest at moderate noise and narrows as ρ grows—consistent with Proposition 1(c)'s prediction that, without variance correction, GPE alone starts to amplify position-correlated noise at high ρ (visible at GDC/CoraFull/ρ=30: 4.87 → 5.59; §5.3 shows that adding variance weighting on the same cell recovers the gain). The per-backbone gain *orders exactly by bias-pathway headroom*, a falsifiable prediction of the decomposition: TARNet (−13.5%), with no internal position absorption, has the most headroom; NetDeconf, GDC, GIAL, and GNUM (−6.2% to −9.6%) already fold partial position signal through representation balancing or treatment-conditioned aggregation; X-learner (−3.4%) has the least, because its cross-fitted pseudo-outcome doubly exploits position—once through neighborhood-aggregated features and again through the counterfactual imputation μ_{1-t}. X-learner's near-neutrality is therefore evidence *for* the dual-root decomposition, not against it: the architecture that most closes the bias pathway gains least from GPE, exactly as Prop 1(a) predicts, and its residual error at high ρ is dominated by path (3), which only VWL can touch (§5.3). The corresponding null—backbone-uniform GPE gains, e.g., X-learner matching TARNet's −13.5%—would have falsified the bias-headroom ordering; the observed 4× spread across backbones confirms it. BNN (S-learner) never benefits, consistent with the single-head argument below.

Table 2 quantifies this across the 90 non-BNN cells and the two independently-generated real-graph benchmarks (BlogCatalog, Flickr). Across the 90 cells, the mean relative PEHE reduction with +GPE is −8.6% (averaged across six non-BNN backbones), and the aggregate effect is significant by Wilcoxon signed-rank ($p < 10^{-5}$) and sign test ($p < 10^{-5}$). Per-backbone reductions span TARNet −13.5%, NetDeconf −9.6%, GDC −9.4%, GIAL −9.2%, GNUM −6.2%, X-learner −3.4%, and BNN +0.9% (worse). The cell-level win counts (Table 2, W/T/L column) concentrate on the five T-learner backbones: 67/75 cells (89%) overall, 56/60 (93%) in the theory-predicted moderate-noise regime ($\rho \leq 20$). On BlogCatalog and Flickr, GPE wins on 3/6 and 4/6 backbones respectively—the BlogCatalog losses cluster on exactly the backbones theory flags as position-redundant (below).

BNN is the only non-beneficiary: as an S-learner it concatenates T into the same representation as X, so GPE's positional signal cannot condition differently for treated vs. control predictions—GPE

Table 1: Rooted PEHE (↓) across $\rho \in \{5, 10, 15, 20, 30\}$. Each cell: mean$^{\pm \text{std}}$ over 5 seeds. Bold = +GPE improves over vanilla. ⋆GDC is current SOTA [14].

		$\rho = 5$			$\rho = 10$			$\rho = 15$			$\rho = 20$			$\rho = 30$		
Model		Cora	DBLP	PubMed	Cora	DBLP	PubMed	Cora	DBLP	PubMed	Cora	DBLP	PubMed	Cora	DBLP	PubMed
TARNet	vanilla	$2.11^{\pm.22}$	$1.88^{\pm.19}$	$2.21^{\pm.10}$	$3.01^{\pm.26}$	$2.47^{\pm.31}$	$3.48^{\pm.25}$	$3.81^{\pm.44}$	$3.62^{\pm.54}$	$4.04^{\pm.62}$	$4.28^{\pm.58}$	$4.21^{\pm.37}$	$3.88^{\pm.34}$	$7.19^{\pm2.46}$	$6.13^{\pm.55}$	$4.19^{\pm.17}$
	+GPE	**$1.87^{\pm.11}$**	**$1.72^{\pm.12}$**	**$1.93^{\pm.12}$**	**$2.52^{\pm.23}$**	**$2.23^{\pm.08}$**	**$3.08^{\pm.31}$**	**$2.98^{\pm.08}$**	**$3.00^{\pm.32}$**	**$3.63^{\pm.31}$**	**$3.41^{\pm.52}$**	**$3.91^{\pm.60}$**	**$3.64^{\pm.16}$**	**$4.68^{\pm.34}$**	**$5.45^{\pm1.20}$**	**$4.04^{\pm.23}$**
GDC⋆	vanilla	$3.57^{\pm.40}$	$3.20^{\pm.23}$	$2.82^{\pm.10}$	$4.03^{\pm.31}$	$3.23^{\pm.34}$	$3.45^{\pm.56}$	$4.19^{\pm.11}$	$3.41^{\pm.28}$	$3.69^{\pm.47}$	$4.36^{\pm.26}$	$3.58^{\pm.23}$	$4.02^{\pm.15}$	$4.87^{\pm.26}$	$4.20^{\pm.39}$	$4.32^{\pm.28}$
	+GPE	**$3.09^{\pm.34}$**	**$2.76^{\pm.31}$**	**$2.56^{\pm.10}$**	**$3.49^{\pm.60}$**	**$2.72^{\pm.11}$**	**$2.93^{\pm.42}$**	**$3.74^{\pm.62}$**	**$2.99^{\pm.17}$**	$3.71^{\pm.73}$	**$3.84^{\pm.22}$**	**$3.15^{\pm.21}$**	**$3.17^{\pm.20}$**	$5.59^{\pm1.71}$	**$3.79^{\pm.19}$**	$4.44^{\pm.70}$
NetDeconf	vanilla	$5.57^{\pm1.68}$	$3.97^{\pm.21}$	$3.22^{\pm.07}$	$6.47^{\pm3.04}$	$3.88^{\pm.16}$	$3.35^{\pm.09}$	$6.55^{\pm1.64}$	$4.12^{\pm.21}$	$4.00^{\pm.43}$	$5.92^{\pm.62}$	$4.26^{\pm.23}$	$4.11^{\pm.43}$	$7.05^{\pm2.21}$	$5.03^{\pm.78}$	$4.53^{\pm.10}$
	+GPE	**$4.93^{\pm.95}$**	**$3.66^{\pm.08}$**	$3.24^{\pm.04}$	**$4.49^{\pm.35}$**	**$3.69^{\pm.16}$**	**$3.30^{\pm.07}$**	**$5.09^{\pm.80}$**	**$3.97^{\pm.13}$**	**$3.67^{\pm.55}$**	**$4.74^{\pm.30}$**	**$4.02^{\pm.15}$**	**$4.02^{\pm.64}$**	**$5.75^{\pm.75}$**	**$4.64^{\pm.43}$**	$4.52^{\pm.36}$
GIAL	vanilla	$5.47^{\pm1.48}$	$4.05^{\pm.22}$	$3.20^{\pm.10}$	$5.94^{\pm.89}$	$4.08^{\pm.11}$	$3.37^{\pm.08}$	$5.80^{\pm1.33}$	$4.57^{\pm.53}$	$3.83^{\pm.36}$	$5.54^{\pm.64}$	$4.35^{\pm.25}$	$4.17^{\pm.17}$	$5.72^{\pm.68}$	$4.71^{\pm.54}$	$4.66^{\pm.27}$
	+GPE	**$4.75^{\pm.53}$**	**$3.81^{\pm.21}$**	$3.22^{\pm.03}$	**$5.52^{\pm1.55}$**	**$3.70^{\pm.10}$**	**$3.35^{\pm.14}$**	**$5.29^{\pm.62}$**	**$4.01^{\pm.25}$**	**$3.32^{\pm.08}$**	**$4.97^{\pm.29}$**	**$4.01^{\pm.35}$**	**$3.61^{\pm.56}$**	**$5.37^{\pm1.07}$**	**$4.21^{\pm.29}$**	**$4.23^{\pm.43}$**
GNUM	vanilla	$6.21^{\pm2.50}$	$3.92^{\pm.08}$	$3.98^{\pm.11}$	$5.64^{\pm1.20}$	$3.90^{\pm.04}$	$4.04^{\pm.12}$	$6.13^{\pm1.58}$	$3.95^{\pm.06}$	$4.29^{\pm.34}$	$5.19^{\pm.73}$	$4.24^{\pm.25}$	$4.27^{\pm.42}$	$5.06^{\pm.69}$	$5.06^{\pm.04}$	$4.21^{\pm.10}$
	+GPE	**$5.04^{\pm.45}$**	**$3.91^{\pm.16}$**	$4.00^{\pm.09}$	**$4.82^{\pm.45}$**	**$3.82^{\pm.05}$**	**$4.01^{\pm.12}$**	**$4.80^{\pm.48}$**	**$3.94^{\pm.10}$**	**$3.94^{\pm.16}$**	**$4.76^{\pm.40}$**	**$4.02^{\pm.05}$**	**$4.11^{\pm.08}$**	**$5.05^{\pm.66}$**	**$4.26^{\pm.13}$**	$4.45^{\pm.59}$
X-learner	vanilla	$2.30^{\pm.18}$	$1.94^{\pm.12}$	$2.12^{\pm.13}$	$3.21^{\pm.49}$	$2.42^{\pm.23}$	$3.11^{\pm.10}$	$3.34^{\pm.11}$	$2.96^{\pm.21}$	$3.70^{\pm.46}$	$4.14^{\pm.57}$	$3.69^{\pm.27}$	$3.51^{\pm.12}$	$4.37^{\pm.53}$	$4.61^{\pm.90}$	$3.93^{\pm.41}$
	+GPE	**$2.22^{\pm.07}$**	**$1.83^{\pm.09}$**	**$1.86^{\pm.10}$**	**$2.55^{\pm.16}$**	**$2.21^{\pm.13}$**	**$3.06^{\pm.19}$**	$3.49^{\pm.41}$	$2.97^{\pm.48}$	$3.72^{\pm.19}$	**$3.64^{\pm.40}$**	$3.70^{\pm.61}$	$3.84^{\pm.13}$	$4.41^{\pm.50}$	**$4.49^{\pm.66}$**	**$3.91^{\pm.10}$**
BNN	vanilla	$5.46^{\pm.46}$	$4.98^{\pm.06}$	$5.55^{\pm.06}$	$5.41^{\pm.45}$	$5.02^{\pm.03}$	$5.44^{\pm.15}$	$5.33^{\pm.42}$	$4.97^{\pm.09}$	$5.44^{\pm.10}$	$5.39^{\pm.40}$	$4.97^{\pm.03}$	$5.41^{\pm.16}$	$5.39^{\pm.46}$	$4.98^{\pm.06}$	$5.32^{\pm.13}$
	+GPE	$5.50^{\pm.46}$	$5.03^{\pm.06}$	$5.54^{\pm.05}$	$5.45^{\pm.44}$	$5.02^{\pm.04}$	$5.46^{\pm.07}$	$5.46^{\pm.41}$	$5.01^{\pm.10}$	$5.48^{\pm.09}$	$5.46^{\pm.42}$	$5.02^{\pm.04}$	$5.44^{\pm.10}$	$5.46^{\pm.47}$	$5.03^{\pm.04}$	$5.44^{\pm.11}$

Table 2: Cross-backbone GPE effect. Mean ΔPEHE (%) and W/T/L are over the 15 (dataset,ρ) cells of Table 1 (negative Δ = improvement). BlogCatalog/Flickr columns: test PEHE, bold = +GPE wins. ⋆GDC is SOTA [14].

	Mean ΔPEHE	W/T/L	BlogCatalog		Flickr	
Model	(%, ↓)	(/15)	vanilla	+GPE	vanilla	+GPE
TARNet	**−13.5**	**15/0/0**	8.03	**7.25**	12.98	**12.13**
GDC⋆	−9.4	12/0/3	3.88	**3.58**	**6.38**	6.86
NetDeconf	−9.6	13/0/2	**3.44**	3.53	6.94	**6.89**
GIAL	−9.2	14/0/1	**3.29**	3.39	**6.27**	6.65
GNUM	−6.2	13/0/2	9.77	**8.81**	13.89	**13.43**
X-learner	−3.4	9/0/6	**5.55**	5.93	9.68	**9.07**
BNN (S-learner)	+0.9	1/1/13	—	—	—	—

Table 3: Rooted PEHE (↓) on DBLP ρ=10 under three community-detection algorithms used by the DGP. Mean$^{\pm \text{std}}$ over 5 seeds. Bold = +GPE improves over vanilla in that column.

Model		Louvain	Leiden	Infomap
TARNet	vanilla	$2.51^{\pm.22}$	$2.72^{\pm.14}$	$2.53^{\pm.24}$
	+GPE	**$2.42^{\pm.37}$**	**$2.48^{\pm.35}$**	**$2.32^{\pm.07}$**
X-learner	vanilla	$2.25^{\pm.15}$	$2.71^{\pm.35}$	$2.53^{\pm.32}$
	+GPE	**$2.18^{\pm.14}$**	**$2.27^{\pm.22}$**	**$2.28^{\pm.09}$**

therefore requires architectures with separate treatment-control heads (TARNet, X-learner, GDC).

The BlogCatalog/Flickr columns of Table 2 are a cross-DGP sanity check: these benchmarks [13] use a different generation protocol (no explicit community bias, no Louvain detection), yet GPE still helps on the majority of backbones, ruling out DGP circularity. The three backbones that flip on BlogCatalog partition cleanly by mechanism: NetDeconf and GIAL employ explicit representation balancing, which on lower-modularity graphs can conflict with GPE's proxy signal rather than complement it; X-learner's cross-fitted pseudo-outcome already exploits position through μ_{1-t}, so GPE's structural signal is redundant (matching its 9/0/6 W/T/L in Table 2). Both flips are predicted by the theory, not counterexamples to it.

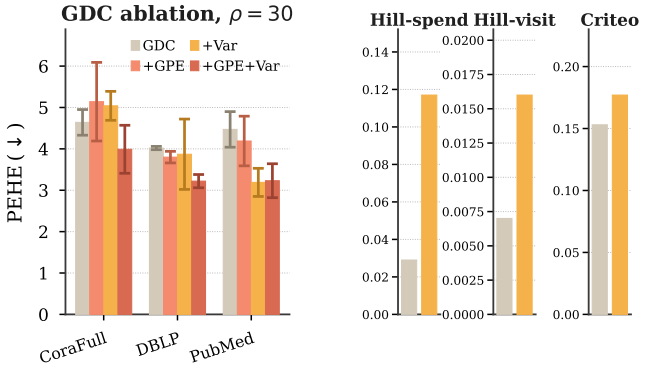

Figure 3: Ablations (mean±std over 5 seeds). Top: GDC at ρ=30. Bottom, three panels: baseline vs. +Var absolute Qini (per-panel y-axis) on the three tabular settings where VWL has dynamic range to exploit; base learner and δ per panel are detailed in §5.4. Arrows show the paired effect; percent lift annotated above.

5.3 Variance Weighting: Complementing GPE

GPE addresses structural bias; variance weighting addresses outcome noise. Figure 3 (left) examines their interaction on GDC at ρ=30 where heteroscedasticity is pronounced; (right) reports Qini on six tabular settings (ablation runs use independent init; small differences from Table 1 reflect seed variation). **VWL on real-graph benchmarks.** On BlogCatalog/Flickr (Guo et al. native low-ρ regime), adding +Var to X-learner yields PEHE 9.68→8.86 on Flickr (−8.5%) and a within-std regression on BlogCatalog (5.55→6.28), confirming VWL generalises beyond our controlled DGP.

At ρ=30, GPE alone *hurts* GDC on CoraFull (4.64→5.14)—positional attention overfits noisy pseudo-labels. CoraFull/DBLP are bias-dominated: the combination wins (3.99, 3.22; up to 20%). PubMed is variance-dominated: VWL alone drives the gain (4.47→3.19, 29%), combination matches (3.23±.41). The two modules target orthogonal error sources per Prop. 1(c); relative importance is dataset-dependent.

5.4 Real-World Tabular Validation

Tabular benchmarks test VWL *as a method*: whether inverse-variance reweighting delivers when heteroscedasticity comes from non-graph sources (revenue variance, treatment-rate imbalance). A positive result supports method scope; §5.3's graph results separately validate the theory. Figure 3 (right) reports relative Qini on three tabular settings, each at its strongest base learner. **Applicability scoping:** we pre-declare scope to settings where the theory can bite—outcomes with non-trivial treatment-effect signal *and* either continuous outcomes or Bernoulli variance with non-degenerate dynamic range (unbalanced μ or non-trivial base Qini). Three binary benchmarks (X5, Lenta, RetailHero) fail this scope (Appendix A): $|\Delta\text{Qini}|<0.005$ at every tuned configuration (3 learners $\times$ 4 δ), consistent with the theory's prediction that variance reweighting cannot refine a near-zero signal—a theory-consistent failure mode, not a method defect.

Within the scoped three:

(i) Heteroscedasticity-dominated continuous outcome (Hillstrom spend). S-learner is weak here (0.029); VWL closes the gap to the stronger T/X-learner baselines with a +303% lift (0.029→0.117, 4×)—the design regime of the learned $\log \sigma^2$ head: continuous residuals with real per-sample noise structure.

(ii) Unbalanced binary with signal (Hillstrom-visit). Bernoulli variance $\sigma^2=\mu(1-\mu)$ is bounded by 1/4 but carries dynamic range when μ is far from 0.5: Hill-visit ($\mu\sim$15%), T-learner base, $\delta=10^{-3}$ gains +129% Qini (0.007→0.016); S-learner is near-random on unbalanced binaries.

(iii) Balanced binary with strong baseline (Criteo). At 85% treated, a T-learner ($\delta=10^{-4}$, weight-clip $[1/3, 3]\times\bar{w}$) gains +16% Qini (0.153→0.177); without clipping the lift drops to +11%—a few near-certain samples dominate gradients under treatment imbalance. (ii) and (iii) have large per-seed std at n=5 and are suggestive.

5.5 Why Does GPE Work? A Mechanism Analysis

A natural question is *how* GPE improves ITE estimation. We test three competing hypotheses through targeted experiments on DBLP and CoraFull (ρ=10, 5 seeds).

H1: Confounding debiasing. Network position is an unobserved confounder in both our DGP and Guo et al.'s [13]—the setting where Veitch et al. [34] prove network embeddings are valid proxy confounders. We measure the propensity head's AUC with/without GPE across 6 (model, dataset) pairs on DBLP and CoraFull: GPE improves AUC in 5/6 pairs, largest on GDC/CoraFull (0.468→0.606) and GDC/DBLP (0.458→0.535)—the same pairs with the largest PEHE reductions. Spearman r_s=0.83 (p=0.04) between ΔAUC and ΔPEHE is the hallmark of confounding adjustment.

H2: Representation balancing—rejected. One might expect GPE to shrink the T=1/T=0 representation gap, as Wasserstein regularization [13] and adversarial balancing [8] do. MMD measurement shows the opposite: GPE *increases* MMD in all 6 comparisons (ratios 1.12–1.49×). GPE preserves treatment-group differences while making them more informative for outcome prediction—not balancing.

Hypothetico-deductive corroboration. Z^* is latent and untestable directly, so Prop. 1 derives three risky predictions with pre-declared nulls, all confirmed on 2,000+ runs: **(B)** GPE gain *orders by bias-pathway headroom*—TARNet −13.5% → X-learner −3.4% (null: uniform gain); **(C)** GPE-alone hurts at large $\bar{\sigma}^2$ where Eq. 2 fails—GDC/CoraFull/ρ=30 4.87→5.59 (null: monotone improvement); **(D)** orthogonal composition by term-dominance—VWL alone −29% on variance-dominated PubMed, joint −20% on bias-dominated CoraFull/DBLP (null: one module dominates). Convergent corroboration is the standard warrant for such latent-cause hypotheses.

6 Discussion and Robustness

Deployment heuristics. Use GPE on position-unaware T-learner backbones when the graph has real community structure (Louvain modularity > 0.3) or topology-correlated treatment; drop it on position-aware backbones where internal proxies of Z^* already saturate the bias reduction. GPE's edge over swapping to Graphformer is *computational*: its $O(n)$ concatenation variant (Appendix B) matches mean PEHE vs. Graphformer's $O(n^2)$ full-graph attention. VWL is most valuable on continuous outcomes with genuine noise heterogeneity; both modules compose when confounding and noise are *both* substantial (Figure 3 top).

Robustness (Appendix B): node2vec, Laplacian, and degree+PageRank all beat no-PE (confirming Prop. 1(a)); Table 3 reruns DBLP under Leiden [33]/Infomap [29] (+GPE wins 6/6).

7 Conclusion, Limitations, and Future Work

Naming *network position* as a dual-root confounder shows prior graph-ITE architectures attack only the bias pathway, leaving path (3) unaddressed; our plug-ins GPE (bias) and VWL (variance) deliver −8.6% mean PEHE (up to −35%) and +303% tabular Qini in the continuous-outcome design regime; on GDC at high noise, VWL alone cuts PEHE by up to 29% on variance-dominated cells and joint GPE+VWL by up to 20% on bias-dominated cells—complementarity per Prop. 1(c), without backbone changes. **Limitations.** Evaluation is semi-synthetic for GPE and tabular for VWL; full-scale Graphormer/GPS, learned PE (LSPE/SignNet), and A/B tests would sharpen the boundary. Prop. 1(a)'s unconfoundedness assumption excludes latent *non-structural* confounders (personality, intent, time-varying preferences). Static node2vec requires periodic recomputation, exposes a cold-start gap, and reduces to noise on near-homogeneous graphs; $O(n^2)$ cross-attention dominates at ≥20K nodes—deploy the $O(n)$ concatenation fusion (§6).

A Supplementary Tabular Benchmarks

Three binary tabular benchmarks—X5 (200K rows), Lenta (690K), and RetailHero (200K)—sit outside the scope of §5.4: all three combine low baseline Qini ($\leq$ 0.015) with μ-rates near 0.5, so neither the treatment-effect signal nor the per-sample Bernoulli variance has the dynamic range that inverse-variance reweighting needs to improve estimation. Table 4 reports vanilla vs. +Var Qini at the default $\delta=10^{-2}$ for the three standard base learners (S, T, X), 5 seeds. Across all 9 (dataset, learner) pairs no configuration improves over baseline by more than 0.005 in absolute Qini, and a broader sweep over $\delta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ and weight-clip thresholds reproduces the same null finding: the best-tuned +Var minus the best

base stays within ±0.005 on each dataset. We read this as the theory's predicted failure mode—variance reweighting cannot refine a treatment effect that is itself near zero.

Table 4: Qini (↑) on the three supplementary binary tabular benchmarks (mean±std over 5 seeds, δ=10^{-2}). Every +Var vs. base difference has $|\Delta|$<0.012 at default config and <0.005 at its best tuned configuration.

	S-learner		T-learner		X-learner	
Dataset	base	+Var	base	+Var	base	+Var
X5	$.013^{\pm.003}$	$.013^{\pm.003}$	$.013^{\pm.004}$	$.014^{\pm.003}$	$.013^{\pm.001}$	$.014^{\pm.002}$
Lenta	$.001^{\pm.007}$	$.001^{\pm.008}$	$.003^{\pm.005}$	$-.002^{\pm.009}$	$.006^{\pm.007}$	$-.006^{\pm.008}$
RetailHero	$.014^{\pm.004}$	$.011^{\pm.005}$	$.012^{\pm.004}$	$.011^{\pm.003}$	$.014^{\pm.004}$	$.015^{\pm.004}$

B Robustness and Design Choices

This appendix collects robustness evidence that was trimmed from the main text for space but supports the paper's core claims.

PE method. On DBLP (ρ=10, 5 seeds), node2vec ($2.37^{\pm.10}$), Laplacian eigenvectors ($2.27^{\pm.16}$), and degree+PageRank ($2.37^{\pm.36}$) all improve over no-PE (PEHE 2.47), confirming Proposition 1(a): the positional signal matters more than the specific encoding method.

Concatenation vs. cross-attention. On DBLP (ρ=10), concatenating node2vec to raw features achieves PEHE $2.10^{\pm.10}$, slightly ahead of GPE cross-attention ($2.32^{\pm.02}$); vanilla scores $2.65^{\pm.14}$. The positional signal itself is the primary driver, and concatenation is the $O(n)$ default recommendation; cross-attention trades a small PEHE premium for 4× lower cross-seed variance.

Graphformer backbone comparison. Replacing the GAT backbone with a sparse Graphformer (learned degree-centrality encoding) and evaluating TARNet on DBLP (ρ=10, 5 seeds): the position-aware backbone alone ($1.95^{\pm.13}$) outperforms GAT ($2.43^{\pm.24}$); GPE substantially improves GAT (2.43 → 2.20, −9%) but *does not* improve the Graphformer (1.95 → 2.06). Any proxy for Z^* helps, whether injected externally (GPE) or learned internally (Graphformer attention).

No-community DGP ablation. With κ_4=0 (no intra-community edge boosting) GPE still improves 4/6 (backbone, dataset) settings, confirming that node2vec captures structural roles (hub vs. peripheral, bridge vs. cluster member) beyond community membership.

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